THE UNIVERSITY OF MELBOURNE

ISYS90049 Digital Business Analysis

Week 11:
Solution Evaluation

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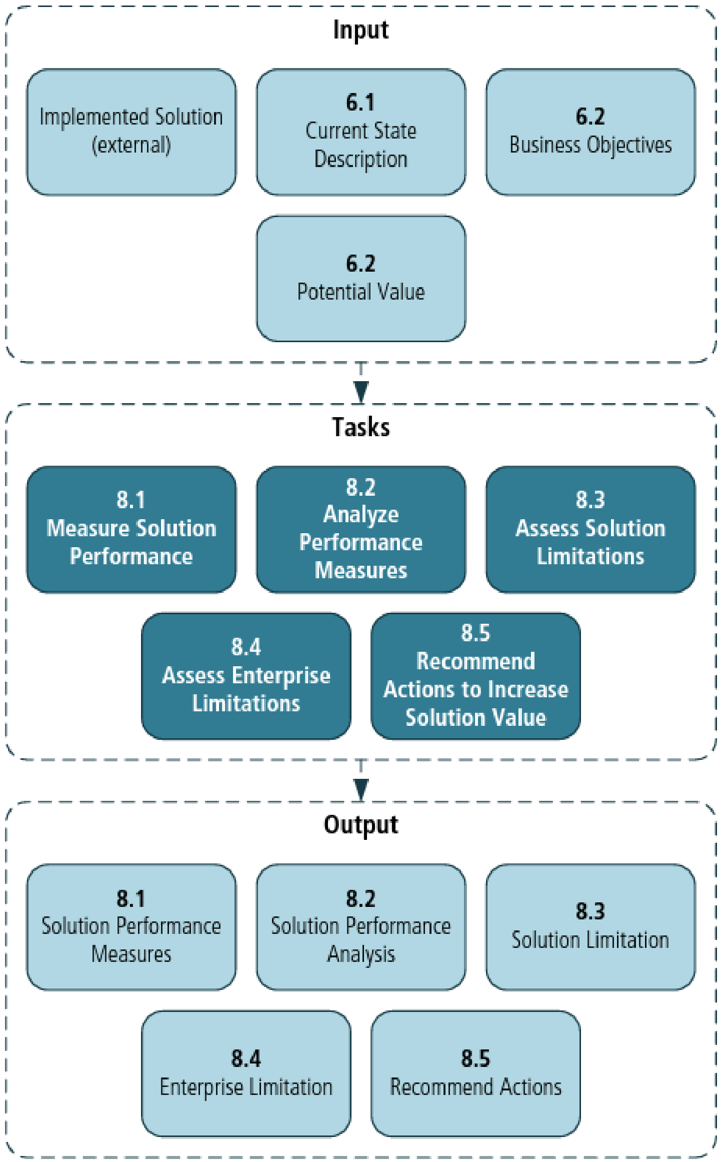
Solution Evaluation

A solution is ready!

Input/output diagram

Figure 8.0.2: Solution Evaluation Input/Output Diagram



BABOK

Knowledge area:
Solution evaluation

Describes tasks to

- assess the performance of and value delivered by an existing solution in the enterprise and
- recommend removal of barriers or constraints that prevent the full realisation of the value

Tasks

Measure solution performance

Analyze performance measures

Assess solution limitations

Assess enterprise limitations

Recommend actions to increase solution value

Tasks

Measure solution performance

Analyze performance measures

Assess solution limitations

Define performance measures and use the data collected to evaluate the effectiveness of a solution

Measure solution performance

Define solution performance measures: ensure performance measures are accurate and relevant. Other performance measures can be elicited. They can be qualitative or quantitative measures.

Validate performance measures

Collect performance measures: consider volume or sample size, frequency and timing, and currency of data

Tasks

Measure solution performance

Analyze performance measures

Assess solution limitations

Recommend

solution value

Provide insights into the performance of a solution

Analyze performance measures

- **Solution performance versus desired value?**
- **Risks to solution performance?**
- **Trends in the data?**
- **Accuracy of performance measures?**
- **Performance variances**: difference between actual and expected performance. Root cause analysis can be necessary to discover the causes of variance

Tasks

Measure solution performance

Analyze performance measures

Assess solution limitations

Determine factors internal to the solution that restrict the full realization of value

Assess solution limitations

- **Identify internal solution component dependencies**: internal dependencies limiting the performance of the entire solution
- **Investigate solution problems**: identify source of the problem (ineffective outputs)
- **Impact assessment**: assess severity of the problem, probability of re-occurrence, impact on the business operations, and capacity of the business to absorb the impact. Identify which problems must be resolved, which can be mitigated and which can be accepted. Additionally, risks to the solution and potential limitations are assessed.

Tasks

Determine how factors external to the solution are restricting the value realization

Assess enterprise limitations

Recommend actions to increase solution value

Assess enterprise limitations

- **Enterprise culture assessment**
- **Stakeholder impact analysis:** Insights into how the solution affects a particular stakeholder group (e.g., functions, locations or concerns)
- **Organisational structure changes**
- **Operational assessment**: determines if an enterprise is able to adapt to or effectively use a solution. Analysts can consider: policies and procedures, processes, capabilities, human resources, tools and technology, etc.

Tasks

Understand the factors that create differences between potential value and actual value, and to recommend a course of action to align them

Assess enterprise limitations

Recommend actions to increase solution value

Recommend actions to increase solution value

- **Adjust solution performance measures**
- **Recommendations**:
  - Do nothing
  - Organizational change
  - Reduce complexity of interfaces
  - Eliminate redundancy
  - Avoid waste
  - Identify additional capabilities
  - Retire the solution

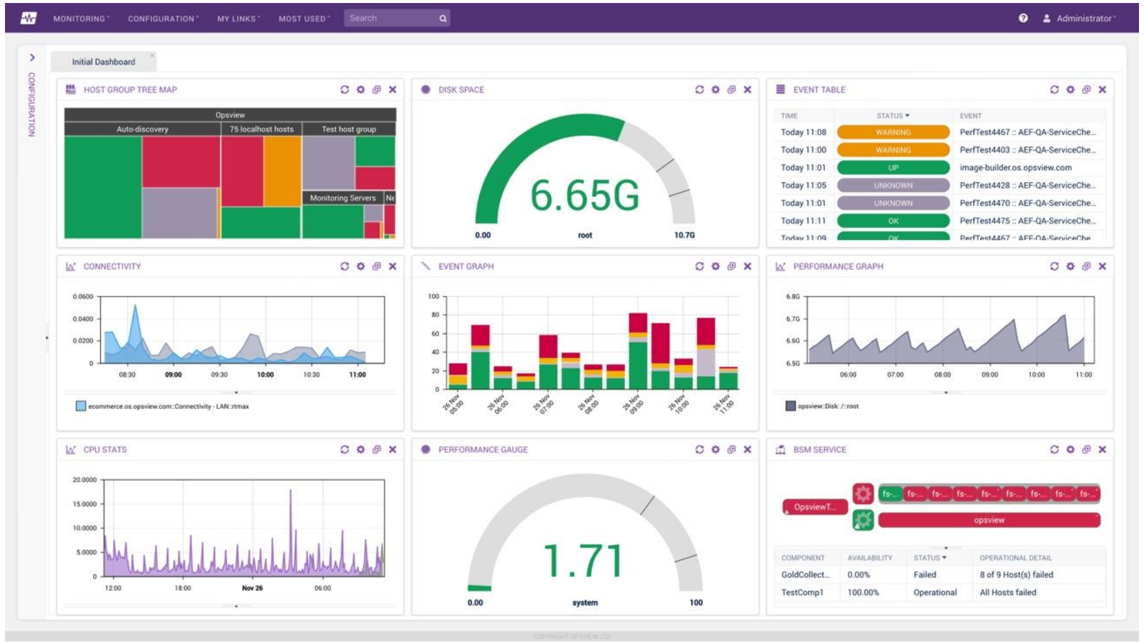
Continuous monitoring

- Evaluation of a solution is not a one-off event
- A metric should be evaluated regularly at appropriate intervals such as weekly or monthly
- Possibilities:

1. **Case 1:** No improvements after one or two intervals
   - Due to delay: identify the reasons and try to remove such delays
   - No benefits being produced: identify if the solution is inefficient or something is missing, then salvage the situation or add missing parts
2. **Case 2:** Progress is going as expected
   - Check if there is anything to improve the benefits or speed up their realisation
3. **Case 3:** Progress is going better than expected
   - Investigate the factors contributing to it

Further analysis

- Additional analysis can be conducted to learn more about how the solution is delivering value
- It might also be the starting point of the next improvement initiative
- Modern data and process mining tools offer dashboards to visualize real-time data and metrics



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Metrics and KPIs

"If you can't measure it, you can't improve it"

- Peter Drucker or Tom DeMarco

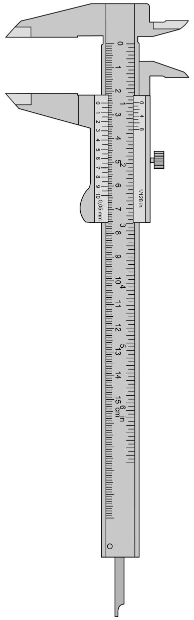
Measures and Metrics

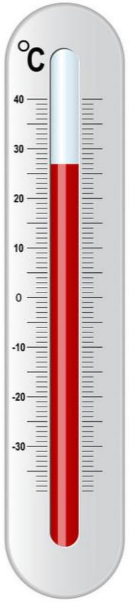
- **Measure** is a fundamental or unit-specific term. For example:
  - Number of customers: 100 customers
  - Number of customers discontinuing the service per month: 10 customers left the service

- **Metric** can be derived from one or more measures. For example:
  - Customer Lifetime Value (CLV): *customer value x average customer lifespan,* e.g. 10$x10 = 100$
  - Churn rate: *customers discontinuing the service/customers,* e.g. 10/100 = 10%

Working with metrics is about …

- Using quantifiable data to analyze the current state:
    1. How are we doing right now?
    2. How do we know this?
    3. Where in this process or operation do we want to improve?
- Working with existing metrics as part of analyzing the current state
- Developing metrics where and when needed





Pre-defined metrics

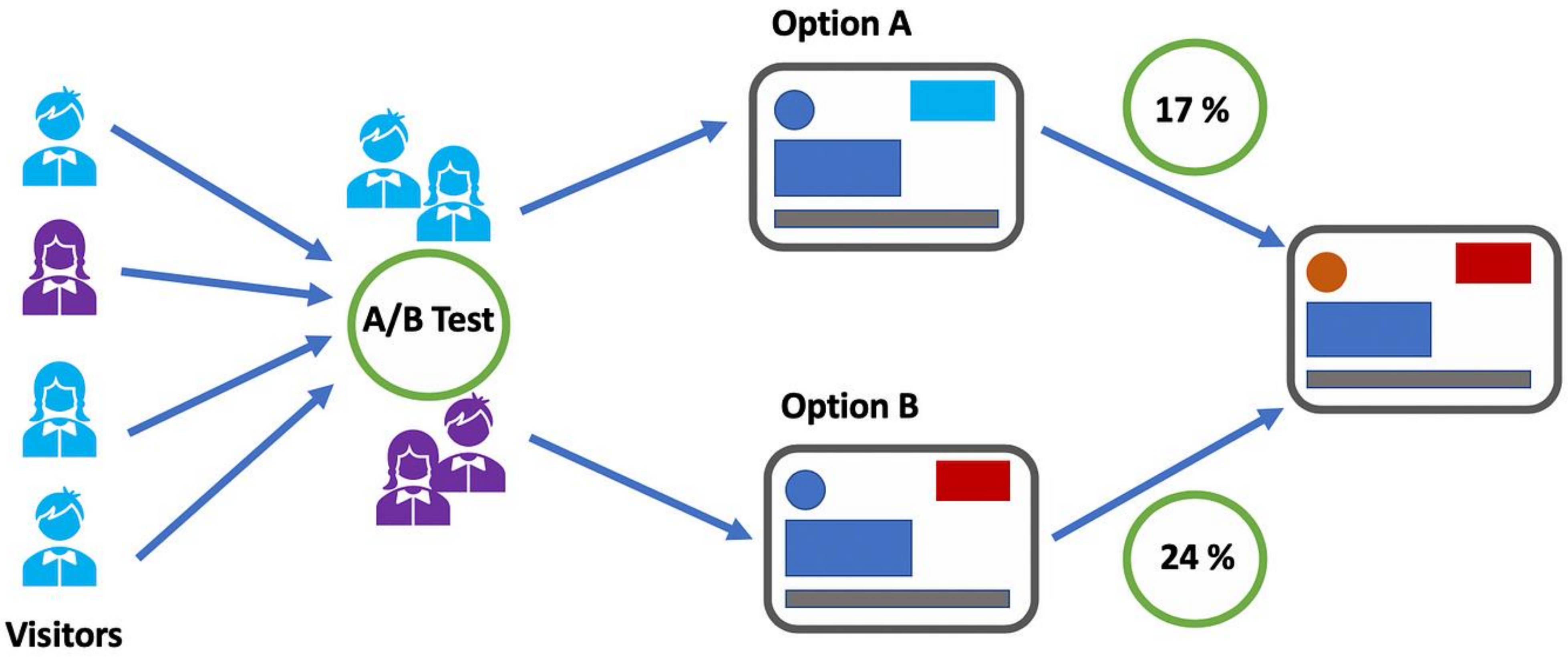
- There are many pre-defined metrics for various industries and processes, but they might
  - not be aligned with the objectives and goals of the company, or
  - organizations have their own strategy, culture, context, processes, capabilities, terminology

Developing a metric

1. Understand the current state and identify areas of improvement
   - In the evaluation, metrics can be used to assess the impact of the solution

2. Implement continuous tracking and monitoring as part of the regular operations of a company
- Ensure there is the capability and set up for monitoring the metric after the project is finished

A/B testing



https://towardsdatascience.com/how-to-conduct-a-b-testing-3076074a8458?gi=515dc4c8a609

Sources of metrics

- **Customers.** As consumers of our products or services, *what customers think or say is important*
  - Is price most important or do they value quality?
  - Is customisation more important than experience?
  - Is the product important or the surrounding services?

- **Problems.** A source of metrics can be where the problems are

- **Objective.** A company might want to achieve something that is not particularly a problem. For example:
  - Increase their level of automation
  - Gain a stronger position in the market
  - Reduce their overall costs

Good business metric characteristics (BABOK)

- **Clear**: precise and unambiguous
  - No room for interpretation as to what the metric measures
- **Relevant**: appropriate to the concern
  - Aligned with the purpose for which the metric is used
- **Economical**: available at a reasonable cost
  - If the costs or the time are high, the metric might be too complex
- **Adequate**: provides a sufficient basis on which to assess performance
- **Quantifiable**: can be independently validated
- **Trustworthy and credible**: based on evidence and research

Ad-hoc metrics – Examples

Selected metrics will depend on the business, industry, function, aspect being measured, etc.

- **Marketing for digital channels**: aims at tracking the performance of marketing efforts on social media, to assess the effectiveness of online campaigns.
  - Some metrics can be: web traffic sources, online incremental sales, social sentiments, Search Engine Optimization (SEO) keyword ranking, and SEO traffic to track, monitor and assess digital marketing

- **Marketing**: aims at increasing sales
  - Some metrics can be: sales growth, sales per product group, average value per purchase, or re-visits

Ad-hoc metrics – Examples (cont.)

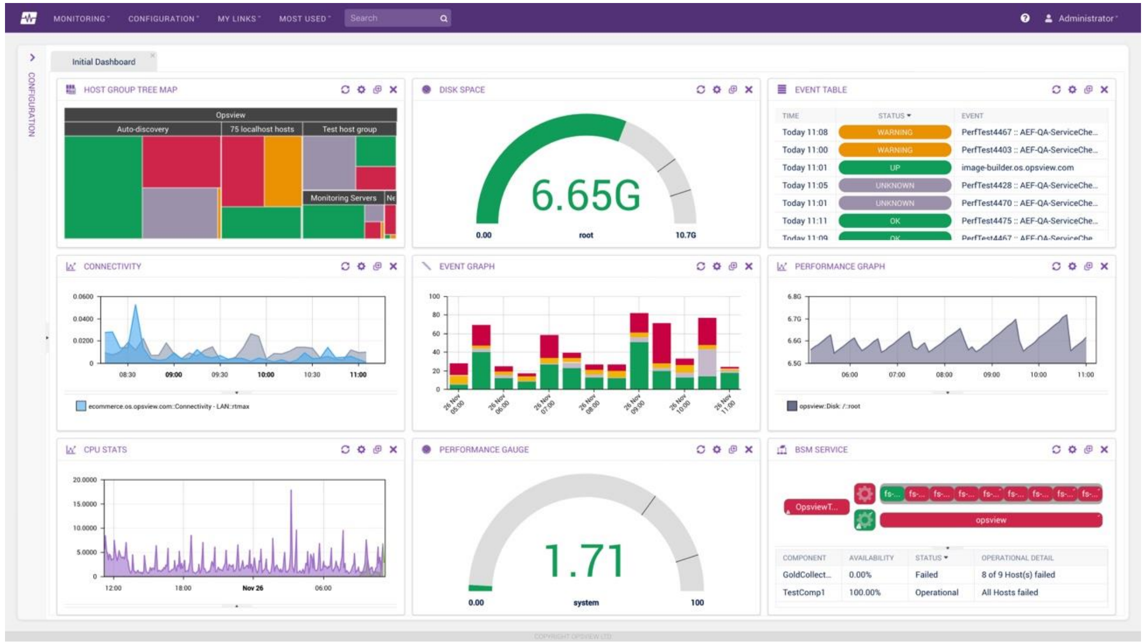
- **SaaS (Software as a Service)**: aims at retaining and attracting new customers
  - Given that they offer subscription-based pricing, some metrics can be: customer retention rate, monthly recurring revenue, customer lifetime value and customer churn rate

- **Social media:** aims at measuring the online presence of a company
  - Some metrics can be: referrals, likes, conversions in social sites, followers, etc.

Are all the metrics key for our project?

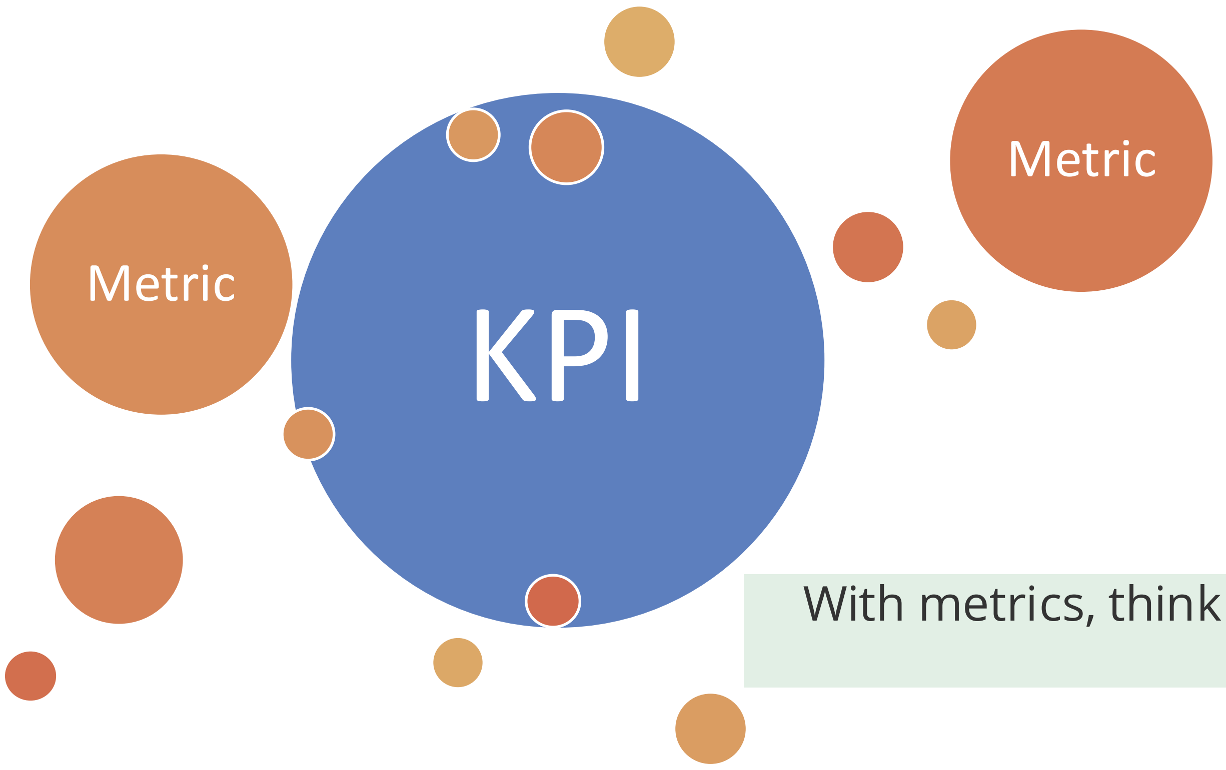
- Metrics track and provide data on your organization’s standard business processes

- Metrics **are not** the most important thing your organization needs to measure, monitor, and perform against to see the progress in your strategic plan



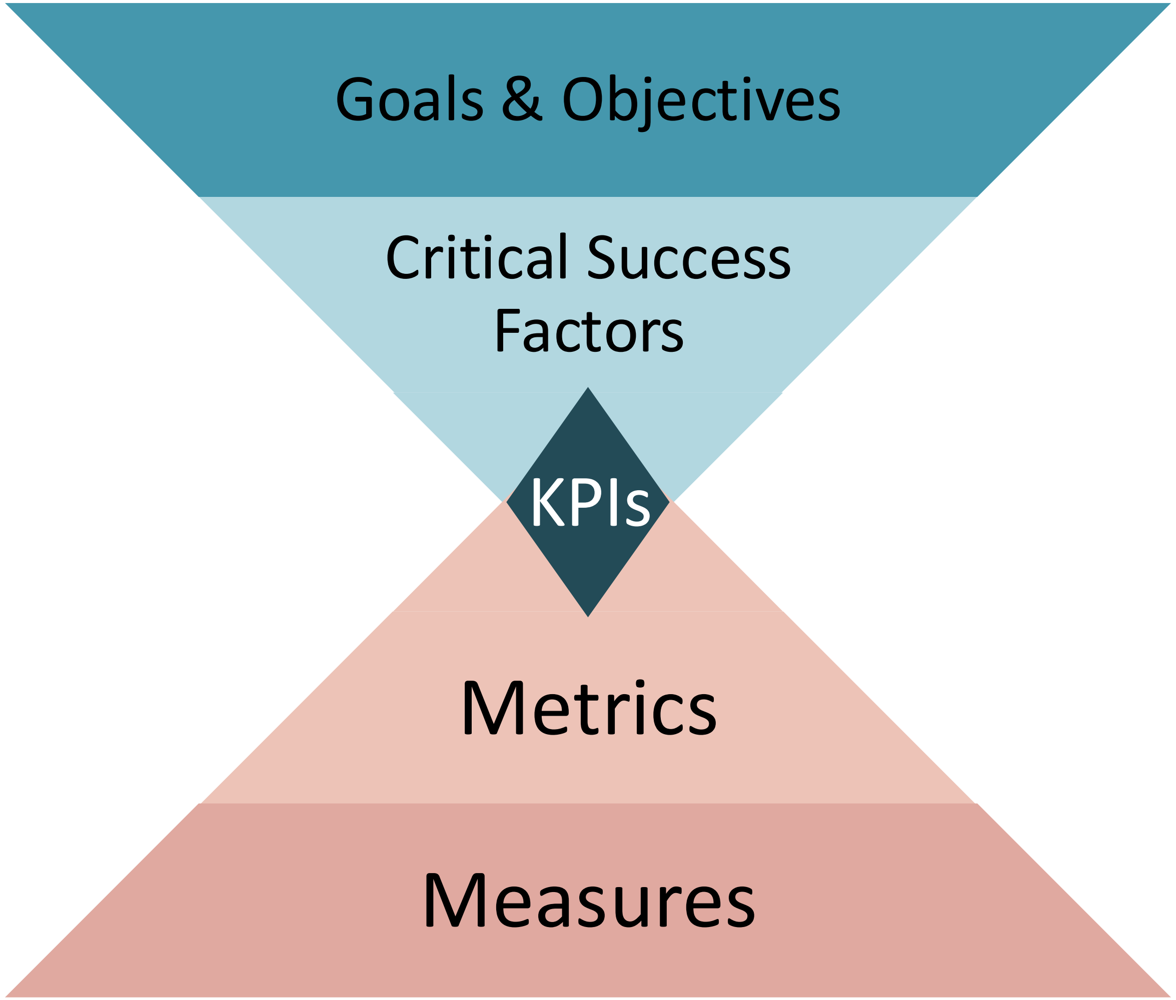
Key Performance Indicator (KPI)

- A KPI is a measurable value that demonstrates how effectively a company is achieving **key business objectives**



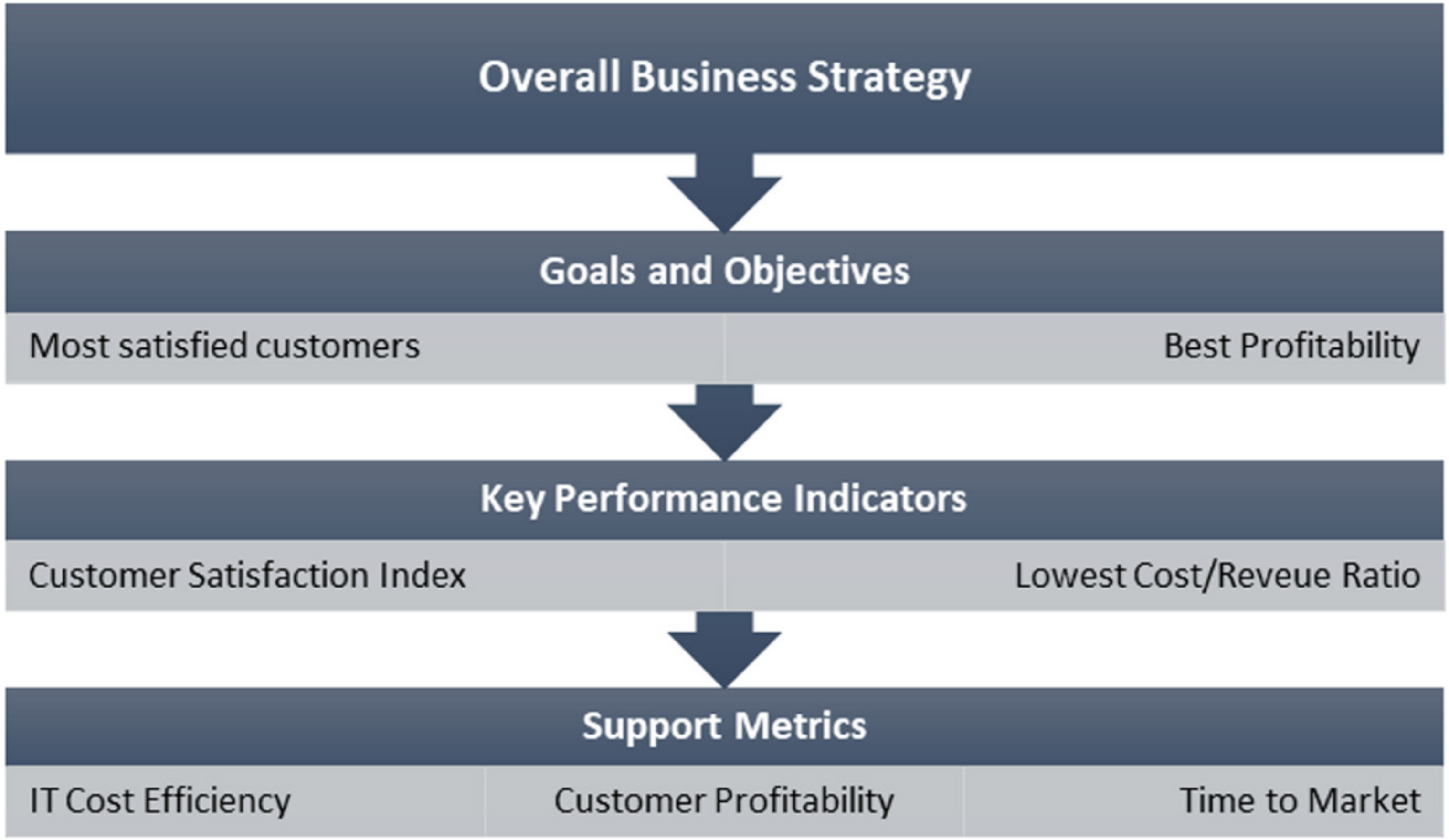
With metrics, think broad. With key performance indicators, think deep.

Metrics and KPIs: The big picture



blog.hyperinnovation.com

Strategy and Metrics



A measurable value that demonstrates how effectively a company is achieving key business objectives

A quantifiable level of an indicator (i.e., a specific numerical measurement) that an organization uses to measure progress

Fig. 10.1 Strategy and metrics

(Milani, 2019)

MyGrocer Example

Review…

Scenario: A family-owned grocery store, ***MyGrocer***, spends many hours every quarterly to perform stock counting and keep track of the expiry date of their stocks. The store has over 1000 different product types (SKUs). They often have to mark down the price of items that are close to their expiry dates. In some cases, they have to dispose all the expired items.

Proposed Solution: A cloud-based retail and inventory system (RIS)

What could be the measures, metrics, and KPIs?

Possible Answers

Measures:

- Number of expired items per month
- Quantity of items marked down due to expiry
- …

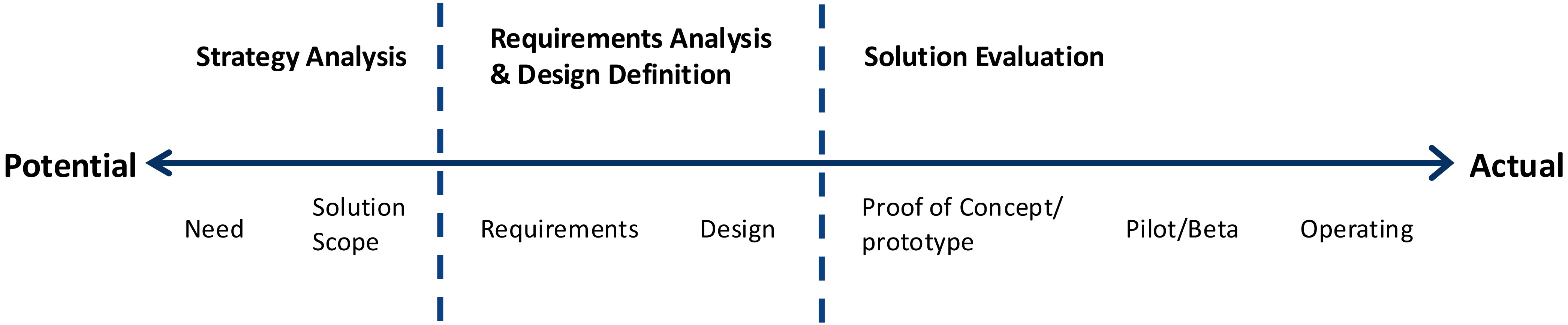
Metrics:

- **% of total stock expired** per quarter = (Expired stock ÷ Total stock) × 100
- **Markdown rate due to expiry** = (Marked-down items ÷ Total items sold) × 100
- …

KPIs:

- Inventory Waste Reduction Rate ((Previous expired value − Current expired value) ÷ Previous expired value × 100)
- …

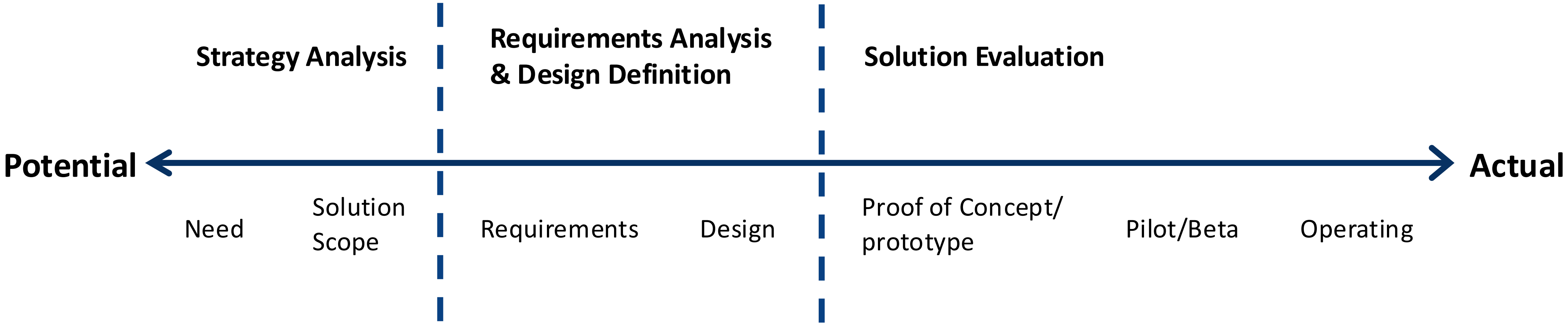
Metrics in Business Analysis Value Spectrum



Metrics and Key Performance Indicators (KPIs) are used to

- Assesses performance of the current state of an enterprise.
- Determine whether the desired future state has been achieved.

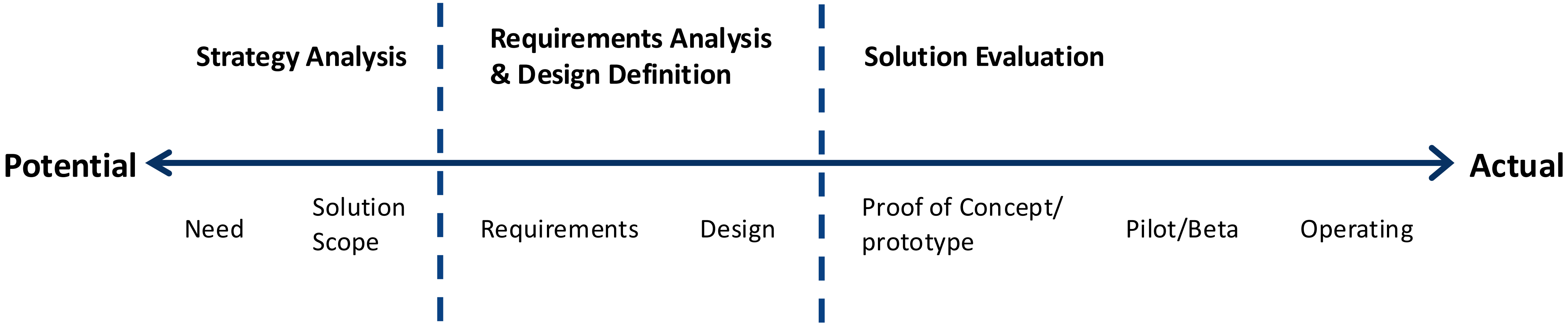
Metrics in Business Analysis Value Spectrum



Metrics and Key Performance Indicators (KPIs) are used to

- Identify how to evaluate the quality of the requirements.
- Select appropriate performance measures for a solution, solution component, or requirement.
- Create and evaluate the measurements used in defining value.

Metrics in Business Analysis Value Spectrum



Metrics and Key Performance Indicators (KPIs) are used to

- Measure solution performance
- Analyze solution performance, especially when judging how well a solution contributes to achieving goals.

A word of caution about metrics and KPIs

Common mistakes with creating metrics and KPIs:

- Measuring the wrong element can lead to irrelevant KPIs
- Wasting resources with dashboards populated with what is available rather than what is useful
- Fixating on data and not on improvements
- Measuring to assess workers' performance
- Brainstorming KPIs rather than creating metrics to reach the goals

Benchmarking

Benchmarking

- Metrics alone do not reflect how a company compares to others

- Benchmarking can be used **to identify performance improvement targets**

Common steps for benchmarking

Choose a process or aspect to compare

Identify the best candidates for the comparison

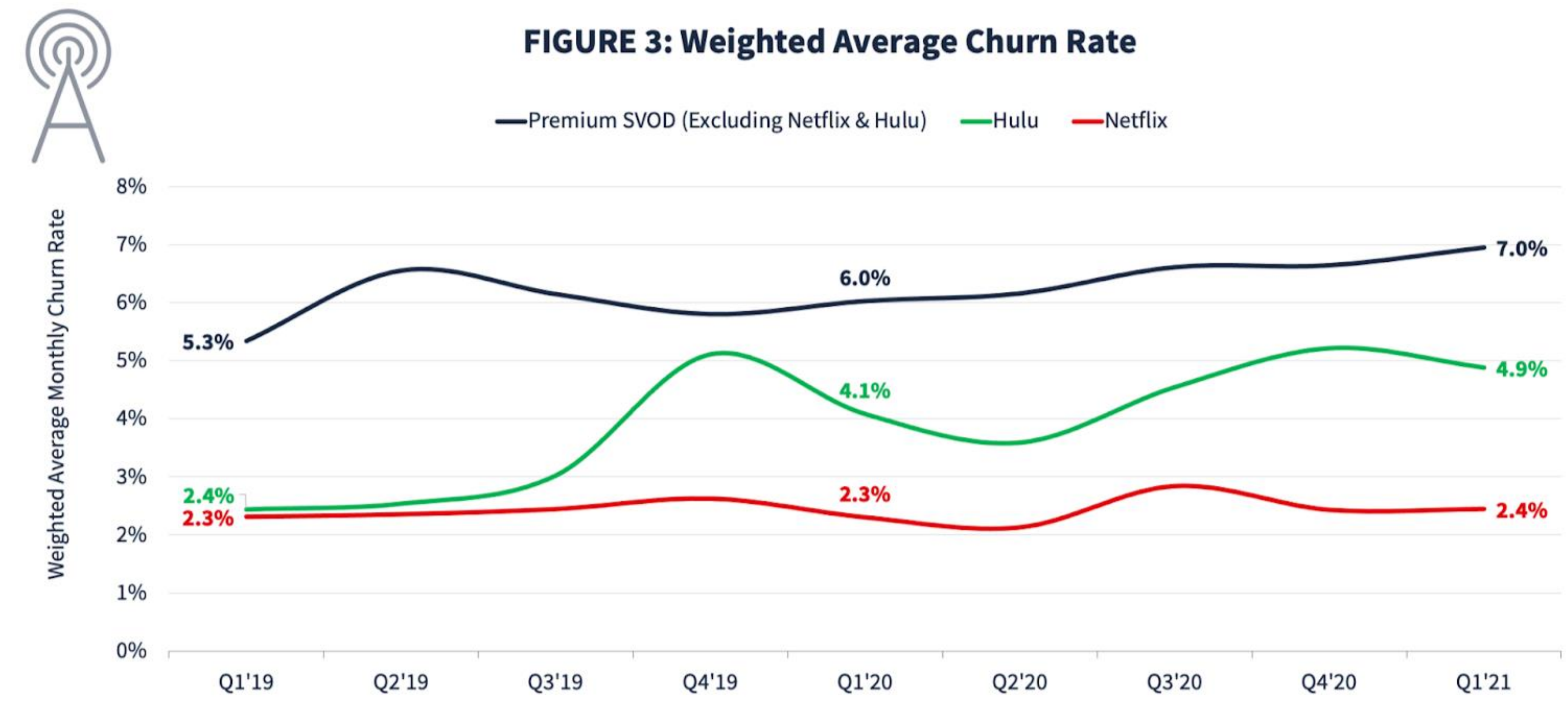
Best performers, leaders or relevant companies

Collect information

Use surveys, collaborations via consultants, and materials available online

Conduct the analysis

Measure the gaps and propose improvements



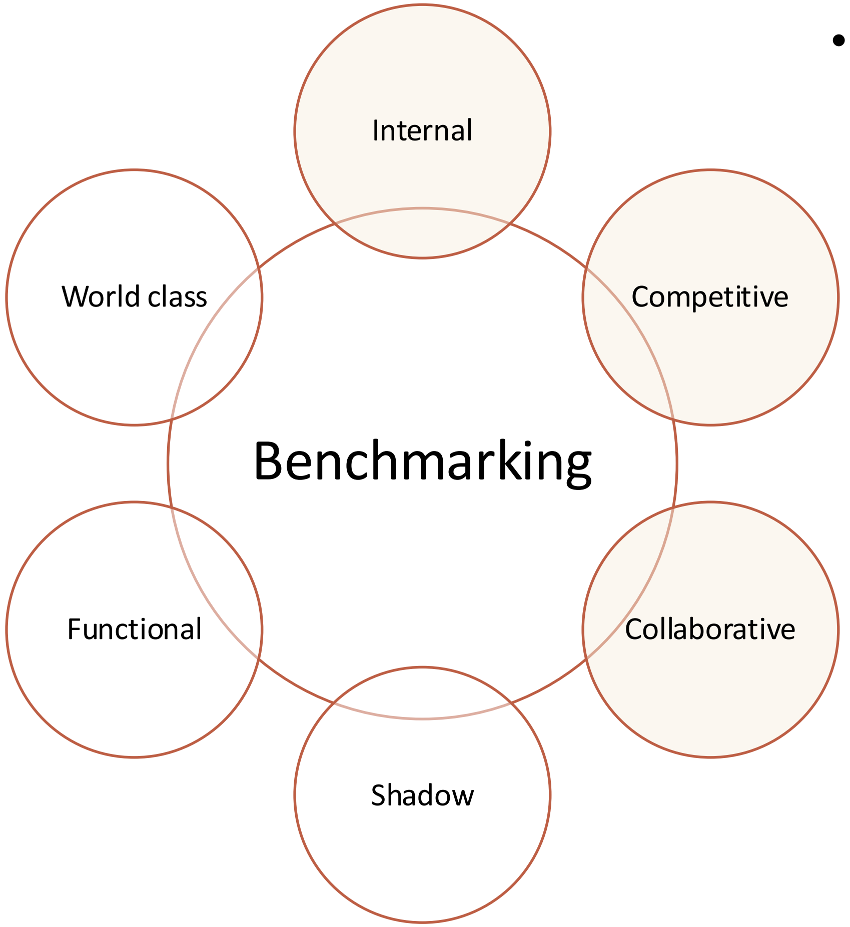
(a) ANTENNA data does not currently include Free Tier Subscribers, MVPD + Telco Distribution, select Bundles (e.g. Hulu/Spotify), and vMVPD Add-ons
(b) U.S. only
(c) Premium SVOD includes Apple TV+, Discovery+, Disney+, HBO Max, Paramount+, Peacock, Showtime, and Starz
(d) Excludes passive churn. A Churn is counted on the day a Subscriber cancels their Subscription, not when the Subscription lapses

Source: Antenna Growth Report 2021

Benchmarking categories

- **Internal**: identify and compare comparable processes within the same organisation (cheap, fast, information is more readily accessible, and has a low risk)

- **Collaborative**: companies within the same industry share primarily quantitative data via industry associations that generate reports

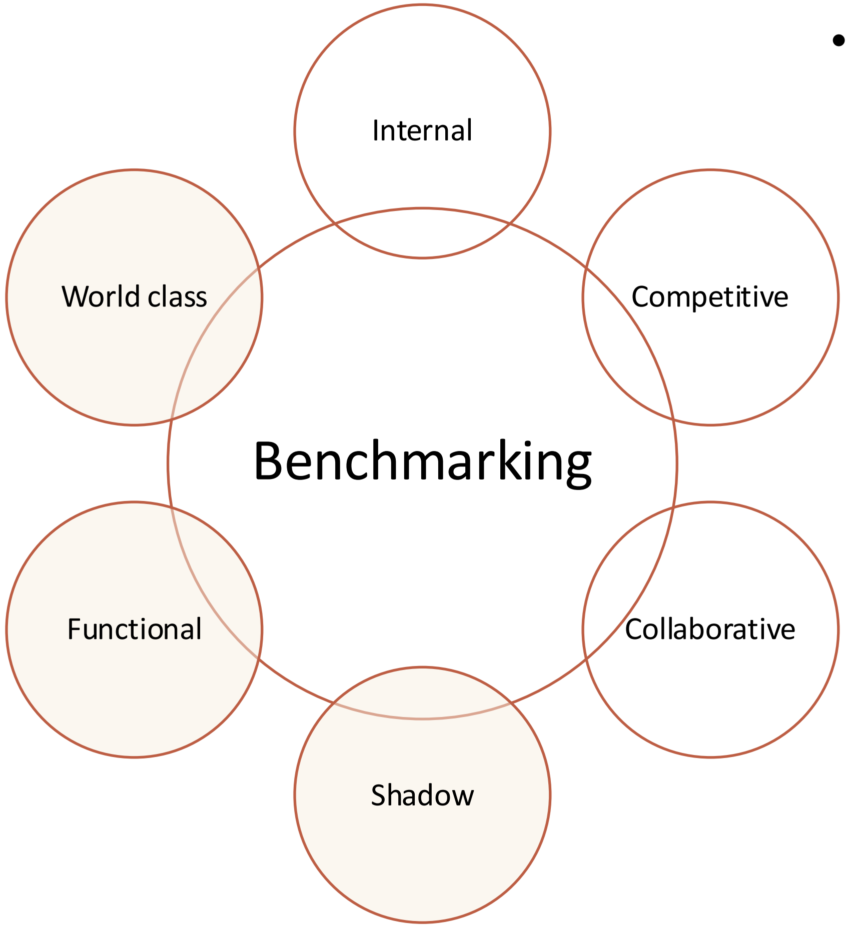


- **Competitive**:
  - *Option 1*: Competitors share data for mutual benefit (Risks: trade secrets being exposed and some companies giving misleading information)
  - *Option 2*: Competitors hire an independent consultancy firm to perform an analysis and generate a report. The report shows the rankings of the companies in regard to different selected parameters but not revealing the company names

Benchmarking categories

- **Shadow**: relies on publicly available data about competitors. Although the data is sometimes incomplete, such methods can provide valuable input

- **Functional**: processes are compared with similar or identical processes from outside the industry
  - A service company might compare its procurement process with that of a manufacturing company



- **World Class**: processes are compared across industries
  - A company operating in manufacturing electronics might compare its billing processes with that of a credit card company